



S10 – Rainfall Hydrographs

The S10 Rainfall Hydrographs apparatus sets out to demonstrate, on a small scale, some of the physical processes found in hydrology.

PRODUCT CODE: S10

Categories: Educational, S Series - Applied Hydraulics & Hydrology, S Series - Hydraulics & Hydrology, S10 - Rainfall Hydrographs

Tags: Bench Mounted , Catchment Rainfall , Floor Standing , Ground Water , Hydraulics & Hydrology , Hydrological Cycle , Impermeable Catchment , Land Drainage , Multiple Storms , Net Rainfall , Previously Saturated Catchment , Rainfall , Rainfall Hydrographs , Reservoir Storage , Runoff Values , S Series , S10 , Single Storms , Surface Recharge , Wells

DESCRIPTION

The S10 Rainfall Hydrographs apparatus sets out to demonstrate, on a small scale, some of the physical processes found in hydrology.

These fall into two related categories:

- The relationship between rainfall and runoff from catchment areas of variable permeability and the abstraction of ground water by wells, with or without surface recharge from rainfall
- The part of the hydrological cycle bounded by the arrival of 'net rainfall' on the ground surface and the catchment runoff by surface streams

Demonstrations are initiated using a gravel filled tank which incorporates facilities for supplying water to the surface of the gravel and measuring the runoff. The gravel tank is manufactured in stove enamelled mild steel supported by a painted mild steel frame.

The equipment can be bench mounted or free standing on a laboratory floor. Water is supplied to two overhead square pattern spray nozzles via a flow control valve, flow meter and solenoid valve. Detachable see-through curtains around the tank contain any spray. Runoff is conducted to an outlet at one end of the tank.

A collection and measuring unit is located near to the outlet from the tank. This comprises a traversing vessel divided internally into 17 storage compartments. The collecting vessel is mounted on a plinth incorporating a motor drive and central drainage trough.

A control console is used to control the traversing vessel and the water supplied to the spray heads. The time each compartment is located under the tank outlet can be preselected and the overall time from the start is displayed.

Water collected in the vessel provides an immediate histogram of runoff as a function of time.

A range of accessories allows demonstrations of surface reservoir retention, depression storage effect and land drainage

These comprise:

- Polythene sheet for impermeable catchment
- Four plastic containers for reservoir storage
- Permeable pipe for tile drain

TECHNICAL SPECIFICATIONS

Tank dimensions

Length: 1.2m

Width: 0.6m

Height: 0.2m

Flow meter range

0.4 – 4.4 litres/min

Runoff collector

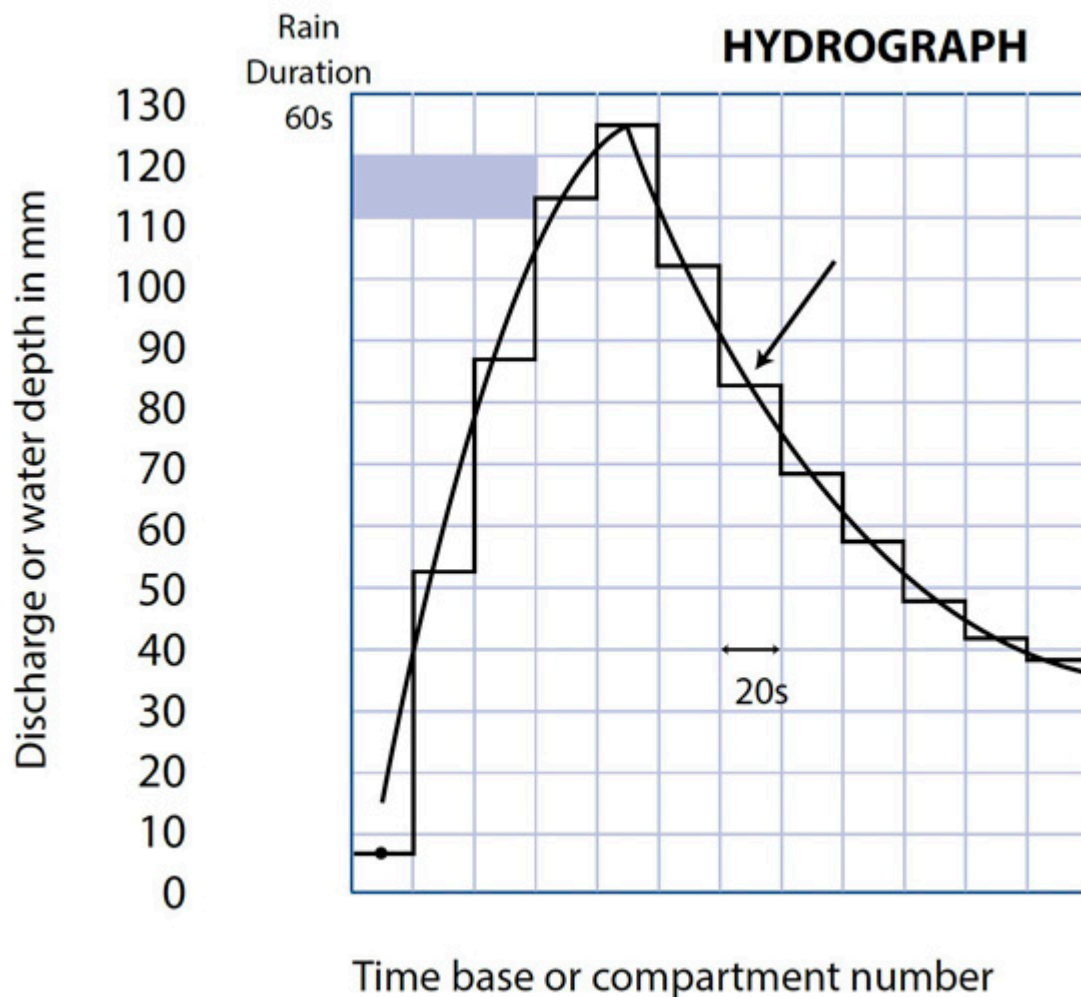
17 x 0.5 litre compartments

FEATURES & BENEFITS

Experimental content

- Storm hydrographs from single or multiple storms
- Storm hydrograph from a previously saturated catchment
- Storm hydrograph for an impermeable catchment
- Storm hydrograph from a catchment with reservoir storage
- Investigate the effect of direction of storm movement on the runoff hydrograph of a catchment
- Investigate the effect of land drainage on the runoff hydrograph of a catchment





S10 Rainfall Hydrograph

Ordering specification

- A unit designed to obtain catchment rainfall and runoff values as functions of time
- Comprising a bench or floor-standing tank with two overhead square pattern spray nozzles supplying water via a flow control valve, flow meter and solenoid valve
- A motor driven traversing vessel with seventeen compartments is moved by a timer beneath the outlet at a preselected rate to collect the runoff and provide an immediate display of the hydrograph
- The tank is 1.2m in length x 0.8m wide x 0.2m deep
- The flow range is 0.4 to 4.4 litres/minute
- A comprehensive user manual is included in the supply

Requirements

Electricity supply:

- **S10-A:** 220-240V/1ph/50Hz
- **S10-B:** 120V/1ph/60Hz
- **S10-G:** 220-240V/1ph/60Hz

Hydraulics bench (F1-10) or cold water supply (4 litres/min) required

Drain

1m³ washed, well graded gravel, range 2.0 – 5.0mm

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

Volume: 1.6m³

Gross Weight: 200kg

Dimensions

Length: 1.58m

Width: 0.9m

Height: 1.05m

Order Codes

- S10-A
- S10-B
- S10-G

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